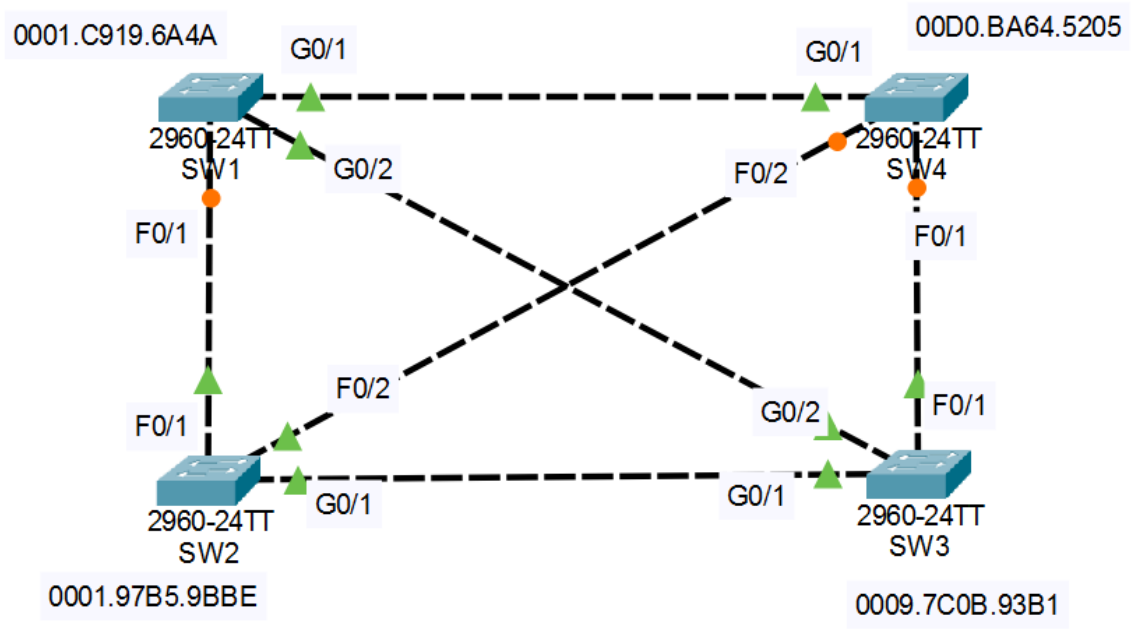
Understanding Spanning Tree Protocol:

Alternate paths in a network are created for network redundancy. To avoid Layer 2 looping while maintaining redundancy, we use Spanning Tree Protocol across switches.



How is STP working in the above topology?

SW2 has the lowest MAC address (lowest bridge priority) = Root Bridge

Root Bridge has all ports designated

```
SW2>en
SW2#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.97B5.9BBE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p
```

Other switches need at least one root port to reach the root bridge. They will have designated ports to forward traffic and, if the link fails, blocked ports to use as alternate paths (network redundancy). The root cost decides them, the neighbour bridge priority and the interface ID of the neighbouring bridge.

The root cost of Fast Ethernet is 19, and for Gigabit Ethernet, it is 4.

And hence, for the rest of the switches –

SW1:

```
SW1>en
SW1#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             Cost        8
             Port        26(GigabitEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.C919.6A4A
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/1            Desg FWD 4         128.25   P2p
Fa0/1            Altn BLK 19        128.1    P2p
Gi0/2            Root FWD 4         128.26   P2p

SW1#
```

SW3:

```
SW3>en
SW3#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0009.7C0B.93B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Root FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p
```

The F0/1 of SW3 is designated because it has lower bridge priority than SW4.

SW4:

```
SW4>en
SW4#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             Cost        12
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.BA64.5205
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Altn BLK 19        128.1    P2p
Fa0/2            Altn BLK 19        128.2    P2p
Gi0/1            Root FWD 4         128.25   P2p
```

Trying different guards in STP –

1. BPDU Guard

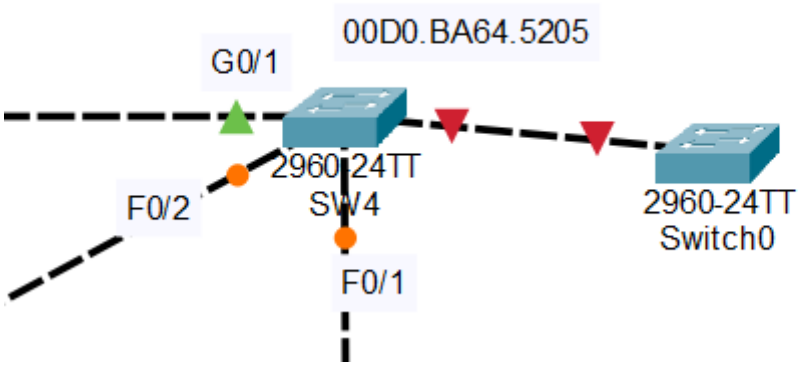
Connecting a PC to SW4 at interface F0/3

```
SW4(config)#int f0/3
SW4(config-if)#spanning-tree portfast default
SW4(config)#
```

The interface F0/3 directly goes from blocking to forwarding state, skipping the listening and learning states and saves time.

```
SW4(config)#int f0/3
SW4(config-if)#spanning-tree bpduguard enable
SW4(config-if)#
```

After enabling BPDU guard, on connecting a switch at the interface F0/3 of SW4, the link is disabled



Then, to get this interface back to normal, we use “shutdown” and “no shutdown” commands in the interface config mode.

2. Root Guard

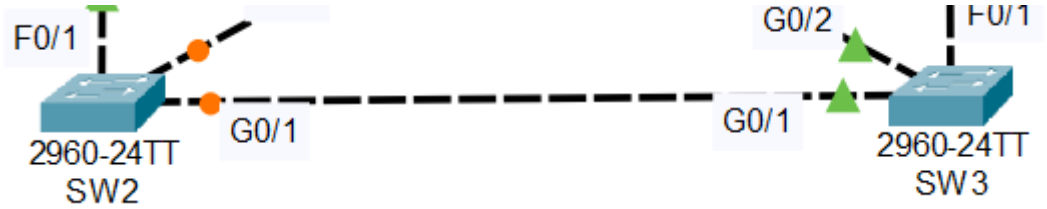
SW3:

```
SW3>en
SW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#spanning-tree vlan 1 priority 4096
SW3(config)#
```

SW2:

```
SW2>en
SW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int g0/1
SW2(config-if)#spanning-tree guard root
SW2(config-if)#%SPANTREE-2-ROOTGUARDBLOCK: Port 0/1 tried to become non-
designated in VLAN 1.

Moved to root-inconsistent state
```



Again

```
SW3(config)#spanning-tree vlan 1 priority 32768
SW3(config)#
```

SW4:

```
SW2(config)#int g0/1
SW2(config-if)#no spanning-tree guard root
SW2(config-if)#
```

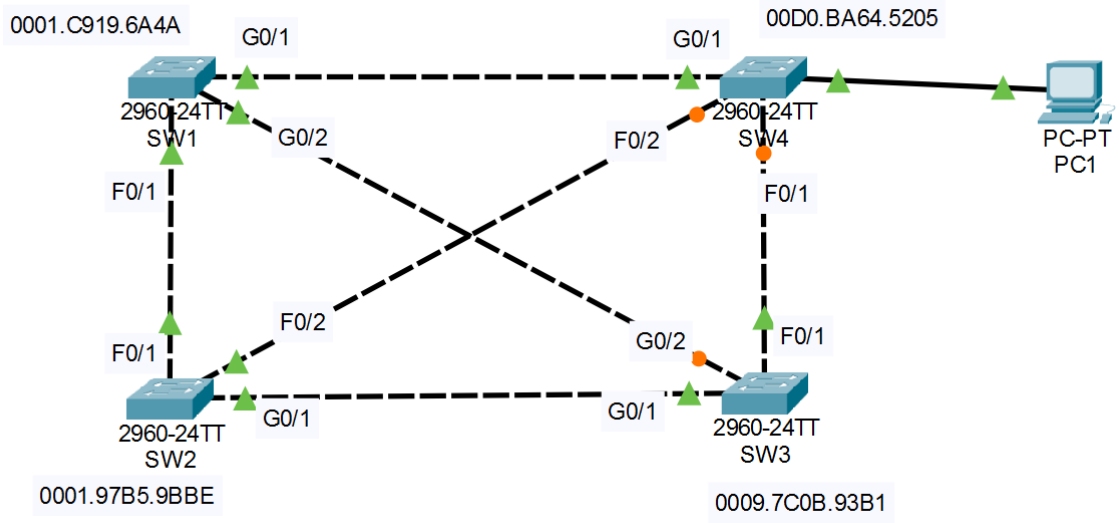
Link restored.

Altering port priority and cost:

1. Changing the root cost of interface f0/1 at SW1

```
SW1(config)#int f0/1
SW1(config-if)#spanning-tree vlan 1 cost 4
SW1(config-if)#
```

Changes in port states



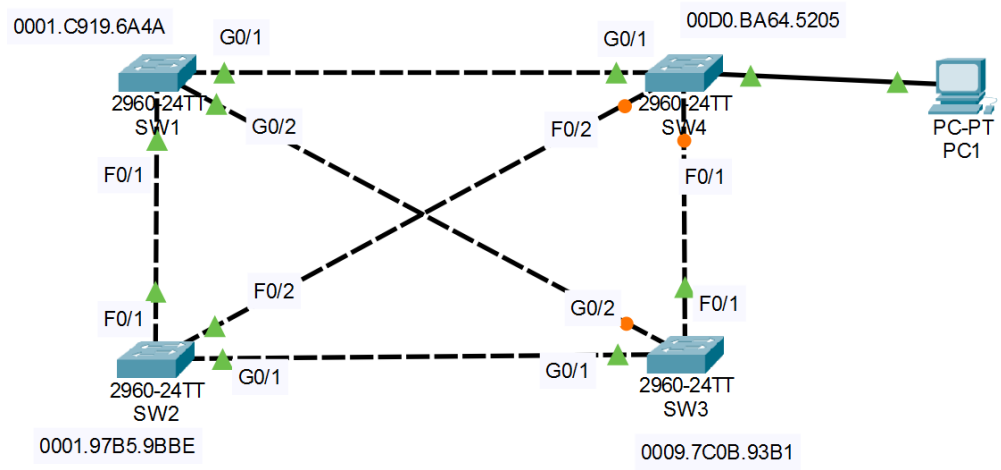
Since now, the best path to the root bridge from SW1 is through F0/1.

2. Trying the tie breaker between two equal paths – the port priority

At SW4

```
SW4(config-if)#int f0/1
SW4(config-if)#spanning-tree vlan 1 cost 4
SW4(config-if)#spanning-tree vlan 1 port-priority ?
  <0-240>  port priority in increments of 16
SW4(config-if)#spanning-tree vlan 1 port-priority
% Incomplete command.
SW4(config-if)#spanning-tree vlan 1 port-priority 16
```

Still the topology is



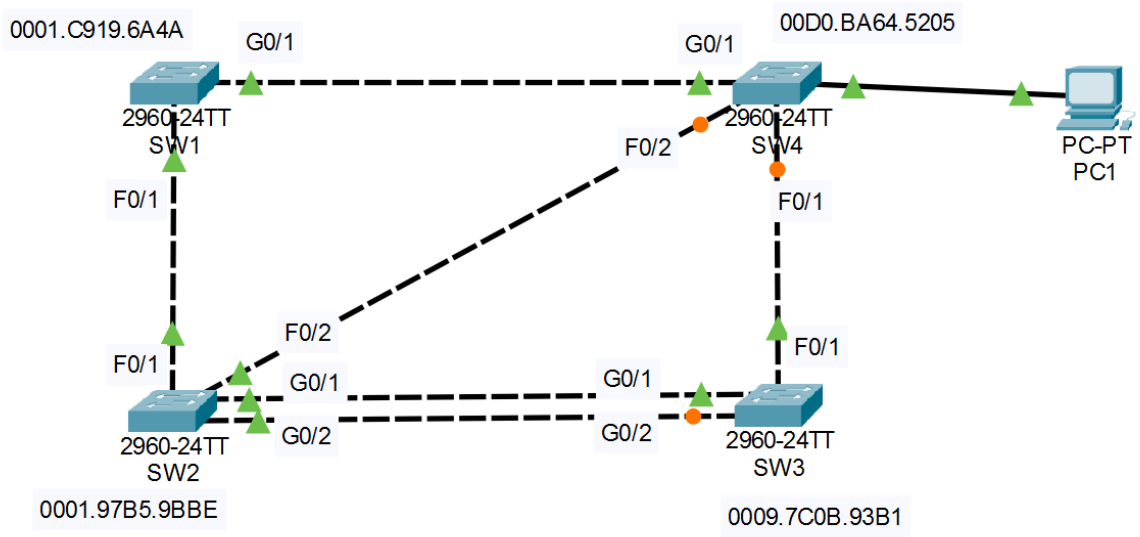
Reason

Even when both the paths, i.e., SW4-SW1-SW2 and SW4-SW3-SW2, have equal costs, the MAC address of SW1 is lower, and that is why that path is chosen.

In Spanning Tree Protocol, the decision order is:

1. Root Bridge ID
2. Path Cost
3. Sender Bridge ID (neighbour switch)
4. Port Priority
5. Port ID

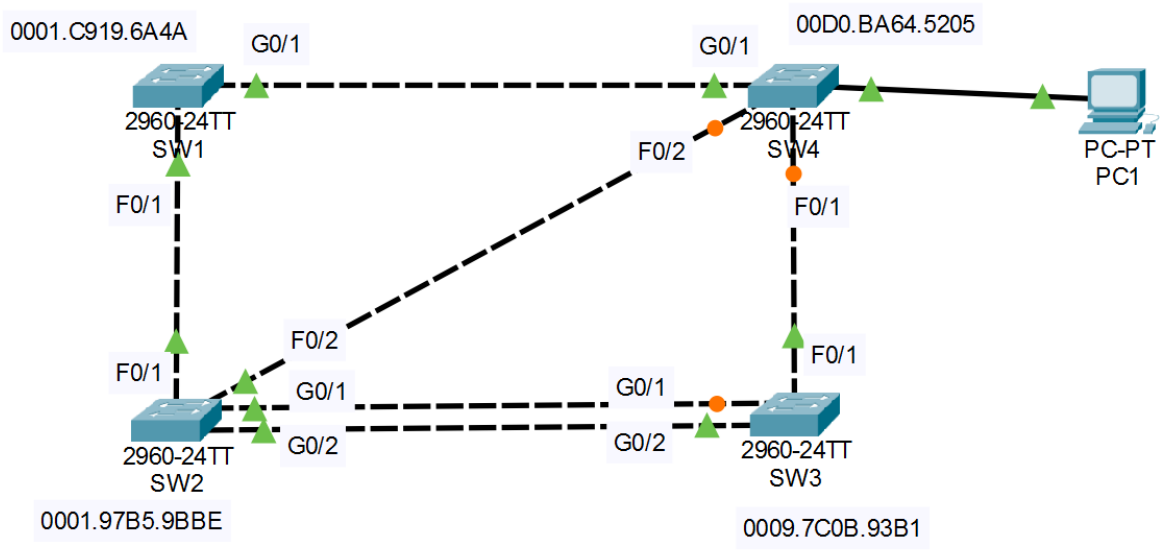
Trying again with a different topology, including parallel connections, to have maximum conditions equal



Changing the port priority on the neighbouring switch (SW2)

```
SW2(config)#int g0/2
SW2(config-if)#spanning-tree vlan 1 port-priority 16
SW2(config-if)#
```

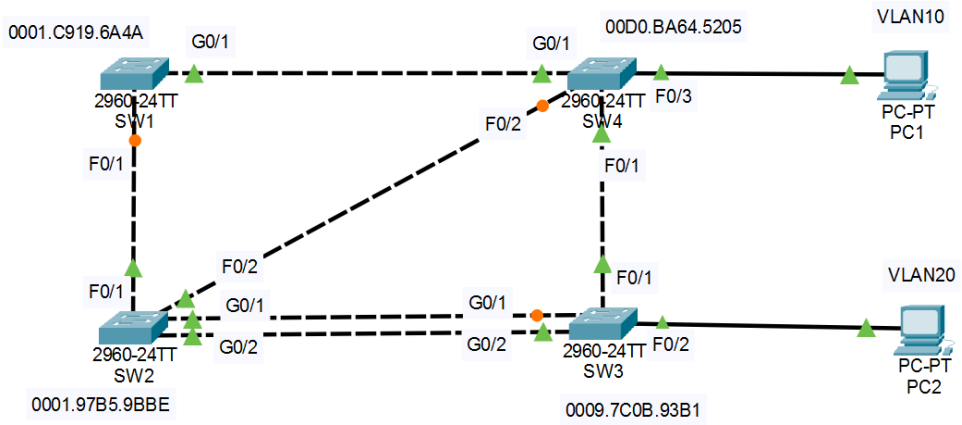
Now the topology changes



Primary and Secondary Routes

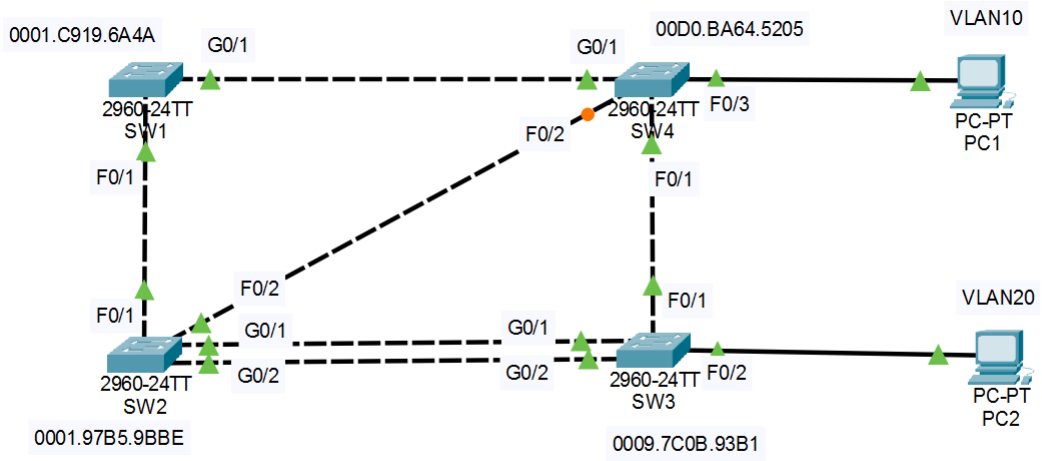
I changed the root cost of F0/1 on SW1 again to 19.

New topology:



Setting SW1 as the root bridge for VLAN 10 and SW2 as the root bridge for VLAN 20.

Final topology:



SW1:

```
SW1(config)#do sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             Cost        12
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.C919.6A4A
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Altn BLK 19        128.1    P2p
Gi0/1            Root FWD 4         128.25   P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0001.C919.6A4A
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0001.C919.6A4A
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0001.97B5.9BBE
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28692  (priority 28672 sys-id-ext 20)
             Address     0001.C919.6A4A
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p
```

SW4:

```
SW4#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             Cost        8
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.BA64.5205
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 4         16.1     P2p
Fa0/2            Altn BLK 19        128.2    P2p
Gi0/1            Desg FWD 4         128.25   P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0001.C919.6A4A
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     00D0.BA64.5205
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/3            Desg FWD 19        128.3    P2p
Gi0/1            Root FWD 4         128.25   P2p
```

SW2:

```
SW2(config)#do sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.97B5.9BBE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         80.25    P2p
Gi0/2            Desg FWD 4         16.26    P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0001.C919.6A4A
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0001.97B5.9BBE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0001.97B5.9BBE
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24596  (priority 24576 sys-id-ext 20)
             Address     0001.97B5.9BBE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p
```

SW3:

```
SW3#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97B5.9BBE
             Cost        4
             Port        26(GigabitEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0009.7C0B.93B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/2            Root FWD 4         128.26   P2p
Gi0/1            Altn BLK 4         128.25   P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0001.97B5.9BBE
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     0009.7C0B.93B1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Gi0/2            Altn BLK 4         128.26   P2p
Gi0/1            Root FWD 4         128.25   P2p
```